

Climate and Tropospheric Oxidizing Capacity

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Keywords

atmosphere, tropospheric chemistry, oxidation, hydroxyl radical, methane

Abstract

The hydroxyl radical (OH) largely controls the tropospheric self-cleansing capacity by reacting with gases harmful to the environment and human health. OH concentrations are determined locally by competing production and loss processes. Lacking strong observational constraints, models differ in how they balance these processes, such that the sign of past and future OH changes is uncertain. In a warmer climate, OH production will increase due to its water vapor dependence, partially offset by faster OH-methane loss. Weather-sensitive emissions will also likely increase, although their net impact on global mean OH depends on the balance between source (nitrogen oxides) and sink (reactive carbon) gases. Lightning activity increases OH, but its response to climate warming is of uncertain sign. To enable confident projections of OH, we recommend efforts to reduce uncertainties in kinetic reactions, in measured and modeled OH, in proxies for past OH concentrations, and in source and sink gas emissions.

- OH is strongly modulated by internal climate variability despite its lifetime of a few seconds at most, with implications for interpreting trends in methane.

- Improved kinetic constraints on key reactions would strengthen confidence in regional and global OH budgets, and in the response of OH to climate change.
- Future OH changes will depend on uncertain and compensating processes involving weather-sensitive chemistry and emissions, plus human choices.
- Technological solutions to climate change will likely impact tropospheric oxidizing capacity and merit further study prior to implementation.

Radical: atom or molecule with at least one unpaired electron in its valence (outer) shell; these highly reactive chemical species drive atmospheric chemistry

Atmospheric lifetime: the average amount of time a molecule remains in the atmosphere once emitted or produced chemically; total atmospheric inventory (burden) divided by total loss rate; can be defined with respect to individual loss pathways; often differs from perturbation lifetime due to nonlinear chemical processes

Radiative forcing: the perturbation in the net radiative flux (W/m^2) at the top of the atmosphere (or tropopause) due to a change in a radiatively active atmospheric constituent, often taken as the difference between the preindustrial and present day; this perturbation will eventually induce a change in Earth's surface temperature to rebalance Earth's energy budget

1. INTRODUCTION

Earth's climate and air quality are strongly coupled to oxidizing photochemical reactions in the troposphere. Tropospheric photochemistry continuously removes reduced gases, as has occurred since oxygen emerged as a major atmospheric constituent. The term atmospheric oxidizing capacity refers to the abundance of major atmospheric oxidants (Thompson 1992). Many challenges exist in quantifying recent trends and variations in this atmospheric cleansing service. For example, it is not possible to directly observe the response of the tropospheric oxidizing capacity to natural climatic fluctuations and to anthropogenic climate change. Current knowledge is thus largely based on proxy evidence.

The hydroxyl radical (OH) (see the sidebar titled Definitions of Chemical Species and Families) is the major tropospheric atmospheric oxidant (Levy 1971) and serves as the focal point for this review. Reaction with OH oxidizes many reduced carbon gases to carbon dioxide (CO_2) and water; such gases include air pollutants, greenhouse gases, and stratospheric ozone-depleting substances. As the primary loss pathway for methane, OH sets the atmospheric lifetime of this greenhouse gas that ranks second behind CO_2 in terms of radiative forcing between the preindustrial and present day (IPCC 2023). Oxidation by OH also sets the lifetime of carbon monoxide (CO), an air pollutant that is a direct product of incomplete combustion and an intermediate produced during the oxidation of methane and many other volatile organic compounds (VOCs). The lifetime of OH is less than a few seconds (Lelieveld et al. 2016), rendering a global observing system impractical if not impossible. Direct measurements of OH, common during field campaigns (Stone et al. 2012), enable process-level evaluations of models (Nicely et al. 2016).

Reactions with CO and methane are key sinks for OH at the global scale, triggering nonlinear chemical feedbacks that cascade across time and space by linking the OH radical with methane and its decadal lifetime (Prather 1996). Complex chemical couplings also occur during the production of OH, which is inextricably tied to tropospheric ozone (**Figure 1**), an air pollutant damaging to human health and vegetation that ranks as the third most important anthropogenic greenhouse gas after CO_2 and methane in terms of radiative forcing from the preindustrial to the present day (IPCC 2023). Primary production of OH, as defined here, requires ozone, ultraviolet radiation, and water vapor. A smaller direct source of OH involves photolysis of VOCs. This OH then initiates the oxidation of VOCs or CO, which produces tropospheric ozone in the presence of nitrogen oxides (NO_x), dwarfing ozone transported from the stratosphere by an order of magnitude while also recycling OH.

Much effort has been expended to quantify and interpret recent decadal trends in methane, but variations due to its sink with OH pose a major uncertainty (Turner et al. 2019). Long-term trends and interannual variability in global mean OH are inferred through inverse methods constrained by observations of methyl chloroform (Lovelock 1977, Montzka et al. 2011, Patra et al. 2021, Prinn et al. 1992), a species primarily lost through reaction with OH. These top-down approaches often disagree with bottom-up approaches with atmospheric chemistry model simulations (Stevenson et al. 2020, Szopa et al. 2021). Conflicting conclusions are highly method

DEFINITIONS OF CHEMICAL SPECIES AND FAMILIES

OH: hydroxyl radical

HO₂: hydroperoxy radical

HO_x: a chemical family defined to account for the continual rapid cycling between the hydroxyl radical and the hydrogen peroxy radical (HO_x ≡ OH + HO₂) as the overall HO_x supply is controlled by slower processes (e.g., primary OH production, formation of stable peroxides) (**Figure 1**)

NO: nitric oxide

NO₂: nitrogen dioxide

NO_x: a chemical family defined to account for the continual rapid cycling between the nitric oxide (NO) radical and the nitrogen dioxide (NO₂) radical (NO_x ≡ NO + NO₂), which occurs through the reactions NO + O₃ → NO₂ + O₂ and NO₂ + *hν* → NO + O followed by O + O₂ + *M* → O₃ + *M*

NO_y: a chemical family consisting of NO_x and its oxidation products

RO₂: organic peroxy radical where R stands for a functional group

ROOR: product of self-reaction or cross-reaction of RO₂

ROOH: organic peroxide

RCHO: organic carbonyl

HCHO: formaldehyde

HOOH: hydrogen peroxide (also H₂O₂)

O³P: “O triplet P” atomic oxygen; ground state of atomic oxygen

O¹D: “O singlet D” atomic oxygen; a higher energy state than O³P

J: used to denote a photolysis rate (1/s)

k: used to denote a kinetic reaction rate coefficient

M: third body (any atmospheric molecule) that removes excess energy released during a reaction

CO: carbon monoxide

CO₂: carbon dioxide

CH₄: methane

VOC: volatile organic compound composed of carbon and hydrogen, and may include oxygen or other elements; includes methane

NMVOC: non-methane VOC

BVOC: biogenic VOC

H₂: molecular hydrogen

CH₃CCl₃: methyl chloroform, a proxy gas used to infer global mean OH

HFCs and HCFCs: hydrofluorocarbons and hydrochlorofluorocarbons; replacement gases for stratospheric ozone-depleting CFCs (chlorofluorocarbons)

dependent (Naus et al. 2019, Turner et al. 2019). This generally underdetermined problem (Prather & Holmes 2017, Turner et al. 2017) has spurred much debate. Here we discuss emerging advances with new satellite retrievals of atmospheric composition and comprehensive airborne sampling strategies that offer promise for progress on this challenging problem.

A key question centers on how competing effects on OH abundance will respond to future changes in climate and anthropogenic emissions, which complicate projections of future oxidative capacity. For example, increased water vapor may enhance OH production, but higher temperatures also increase the rate of OH destruction via reaction with methane. Calculation of global mean OH requires integrating over local OH, which in turn is controlled by various processes that together set the lifetimes of many trace gases directly relevant to climate, air quality, the

Primary production

of OH: ozone photolyzes to produce the O¹D radical, followed by reaction of O¹D with water vapor to yield two OH molecules

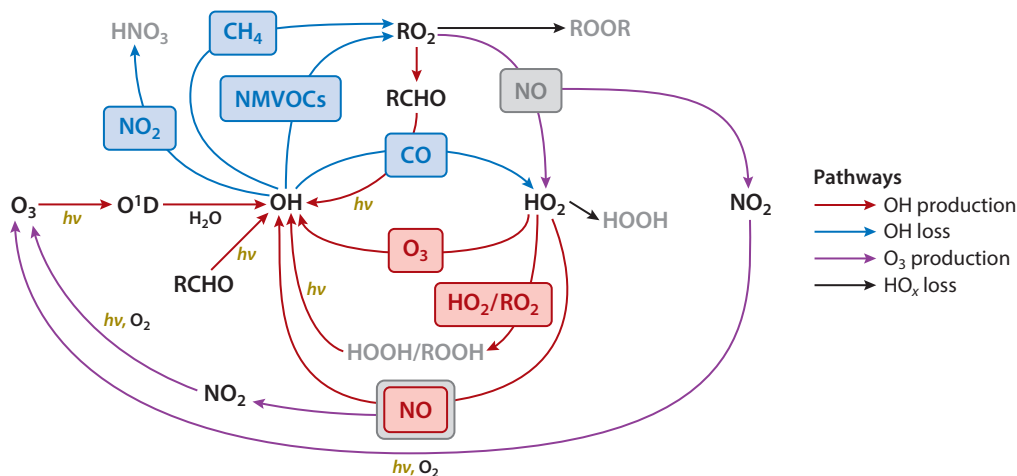


Figure 1

Simplified schematic of OH chemistry. Species in light gray font are radical sinks if they deposit or are otherwise lost without further oxidant regeneration. NO is highlighted in gray to further denote its role in the rate-limiting reactions that control tropospheric ozone production. Abbreviation: NMVOC, non-methane volatile organic compound.

Top-down approach: inverse modeling approach that uses an observed quantity and a known relationship to the variable of interest to infer that variable

Bottom-up approach: forward modeling approach that simulates processes controlling the variable of interest

Photochemical box model: zero-dimensional model of atmospheric chemistry; usually includes more complex chemistry than is possible to represent in three-dimensional numerical models of global atmospheric chemistry and transport due to computational expense

stratospheric ozone layer, and thus public health and ecosystem welfare. Here we consider regional and temporal variations in OH sources and sinks. Although much of our emphasis is placed on the modern atmosphere, we discuss processes controlling OH abundances during preindustrial and glacial times, which can provide clues as to how OH will respond to future global change.

2. OH IN THE MODERN ATMOSPHERE

Knowledge of the OH budget globally is largely based on atmospheric chemistry models that include photochemical reactions measured in lab studies (e.g., Burkholder et al. 2019) or in some cases based on theory, and are evaluated with field measurements (e.g., Miller & Brune 2022). Global atmospheric chemistry models disagree as to the importance of different OH production and loss terms (Wild et al. 2020). Advancing understanding requires observational evidence that can discriminate across models and test their process representation. A typical approach to model evaluation uses field observations of the species involved in OH production and loss to constrain photochemical box models. These studies are limited to relatively small regions and time periods covered by field campaigns (Stone et al. 2012), which sample the local processes over which the global mean integrates. A synthesis of two decades of field campaigns with varying environmental and chemical conditions found that simulations of free tropospheric OH generally agree with observations within the measurement uncertainty, while pointing to a need for further study of regions with plentiful reactive carbon compounds (Miller & Brune 2022). Long-term measurement sites have also yielded foundational, process-based insights into OH drivers (e.g., Rohrer & Berresheim 2006). (For definitions of the chemical species discussed below, see the sidebar titled Definitions of Chemical Species and Families.)

2.1. Budget

Given its short lifetime [at most a few seconds (Lelieveld et al. 2016)], OH rapidly achieves steady state between production and loss pathways (**Figure 1**). The contributions of different source and sink terms to the OH budget vary in space and time, as illustrated in **Figure 2**. On average

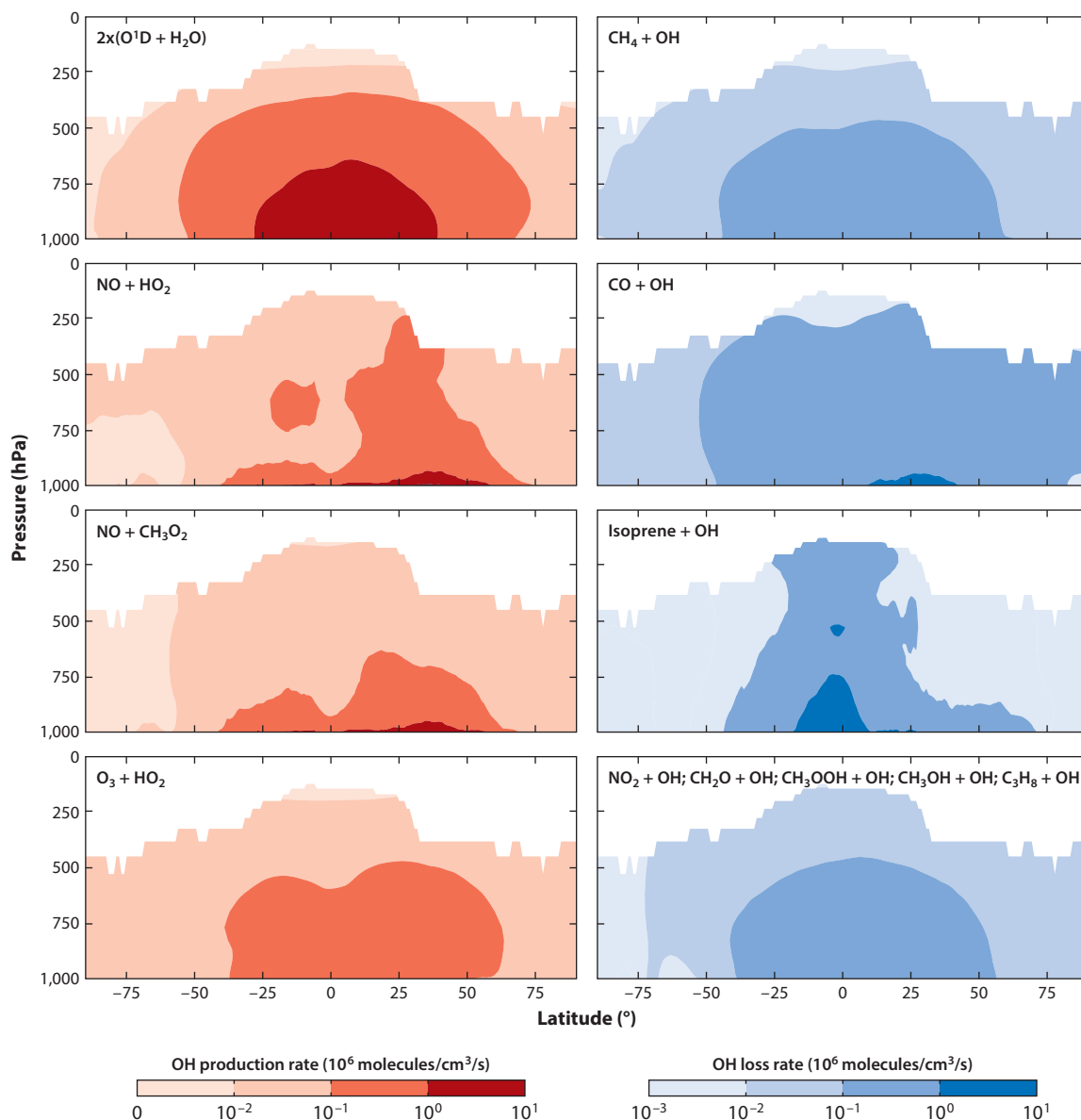


Figure 2

Zonal mean distributions of annual mean OH production (*left panel*) and loss (*right panel*) pathways depend on the spatial patterns of source and sink gases as well as temperature-dependent rate coefficients (**Table 1**). This illustrative example was produced from an initial-condition ensemble produced with the Community Earth System Model version 2–Whole Atmosphere Community Climate Model version 6 chemistry-climate model for year 2005 conditions for greenhouse gas concentrations, air pollutant emissions, and other forcings (Fiore et al. 2022). The OH production pathways (from top to bottom) include $2\times(\text{O}^1\text{D} + \text{H}_2\text{O})$, which produces two OH molecules each time the reaction proceeds; $\text{NO} + \text{HO}_2$; $\text{NO} + \text{CH}_3\text{O}_2$; and $\text{O}_3 + \text{HO}_2$. The OH loss pathways (from top to bottom) include the OH reactions with CH_4 , CO, isoprene, and other trace gases: NO_2 , formaldehyde (CH_2O), methyl hydroperoxide (CH_3OOH), methanol (CH_3OH), and propane (C_3H_8).

Steady state:

a balance between production and loss of a chemical species such that there is no change in concentration with time; pseudo-steady state occurs for very short-lived species such as OH where the processes controlling its production and loss change on much slower timescales than the OH lifetime

globally, primary production of OH dominates (Chua et al. 2023, Lelieveld et al. 2016, Zhao et al. 2020), with zonal, annual mean contributions from this pathway to total OH production ranging from 20% to 80% (Anderson et al. 2021, Spivakovsky et al. 2000) (**Figure 2**). Primary production, which controls the total supply of HO_x in the troposphere, is closely coupled to climate through its dependence on both water vapor and ultraviolet radiation ($h\nu < 340$ nm) and thus maximizes in the lower tropical troposphere (**Figure 2**). Multiple secondary production pathways recycle OH consumed during the initial oxidation reaction of a reactive carbon molecule (e.g., CO + OH or VOC + OH), yielding HO₂, followed by reaction with NO, ozone, another HO₂, or other RO₂ radicals. Additional cycling may occur through peroxides (HOOH, ROOH) or carbonyls (RCHO) that can photolyze to produce OH (**Figure 1**). Lumping several terms from **Figure 1** into the category other, we can express the total OH production rate as (see also the sidebar titled Definitions of Chemical Species and Families)

$$P_{\text{OH}} = \frac{2J_{\text{O}_3 \rightarrow \text{O}(^1\text{D})}k_{\text{O}(^1\text{D})+\text{H}_2\text{O}}[\text{O}_3][\text{H}_2\text{O}]}{k_{\text{O}(^1\text{D})+\text{M}}[\text{M}] + k_{\text{O}(^1\text{D})+\text{H}_2\text{O}}[\text{H}_2\text{O}]} + k_{\text{HO}_2+\text{NO}}[\text{HO}_2][\text{NO}] + k_{\text{HO}_2+\text{O}_3}[\text{HO}_2][\text{O}_3] + J_{\text{HOOH}}[\text{HOOH}] + \text{other}, \quad 1.$$

where steady state for O¹D is assumed in the first term on the right-hand side. **Figure 2** shows that the balance of these recycling reactions varies with atmospheric region, reflecting the location of source or sink gases, the distribution of solar radiation, and temperature-sensitive reaction kinetics. For example, NO + HO₂ is largest where NO_x emissions peak at northern mid-latitudes (Gaudel et al. 2020) and increases in relative importance with altitude, reflecting both a faster reaction at lower temperatures (**Table 1**) and less efficient primary production in the dry upper troposphere.

The largest sink for tropospheric OH globally is reaction with CO at ~40%. Reactions with methane (~10%) and non-methane volatile organic compounds (NMVOCs) (~20–30%) serve as other key sinks for OH (Chua et al. 2023, Lelieveld et al. 2016, Spivakovsky et al. 2000). The NMVOC sink may include carbonyls formed from reactions of HO₂ or RO₂ with RO₂, which can in turn amplify primary production of OH through subsequent photolysis (Jaeglé et al. 2001). Peroxides serve as permanent radical sinks if removed by wet or dry deposition, by further reaction with OH, or by photolysis pathways that do not recycle OH. Including smaller loss terms in the category other (e.g., OH reaction with reactive nitrogen and sulfur, halogens, and other species), the total OH loss rate can be expressed as

$$L_{\text{OH}} = k_{\text{CO}+\text{OH}}[\text{CO}][\text{OH}] + k_{\text{CH}_4+\text{OH}}[\text{CH}_4][\text{OH}] + \sum_{i=1}^N k_{\text{NMVOC}_i+\text{OH}}[\text{NMVOC}_i][\text{OH}] + k_{\text{HOOH}+\text{OH}}[\text{HOOH}][\text{OH}] + k_{\text{HO}_2+\text{OH}}[\text{HO}_2][\text{OH}] + \text{other}. \quad 2.$$

Atmospheric chemistry controls which regions are most important to the removal of reactive carbon species. In **Figure 2**, the patterns of CH₄ + OH versus CO + OH are shaped by the strong temperature dependence of CH₄ + OH and the lack thereof for CO + OH (**Table 1**). In contrast to methane and CO, with their atmospheric lifetimes of a decade and a few months, respectively, the OH sink to isoprene, the most abundant individual biogenic volatile organic compound (BVOC) with a lifetime as short as less than 1 h, is largest in tropical and summertime mid-latitude source regions (especially the southeastern United States) (Guenther et al. 2012) (**Figure 2**). **Figure 2** shows evidence of convective lofting of isoprene to the tropical mid-to-upper troposphere where its reaction with OH proceeds faster at lower temperatures (**Table 1**). Myriad diverse OH sinks pose large uncertainties in local- to regional-scale OH concentrations, although advanced measurement techniques point to a key role for oxygenated VOCs in regions such as the Amazon with high biogenic emissions (Pfannerstill et al. 2021).

Table 1 Temperature-dependent kinetic reaction rate coefficients (*k*) determining OH production and loss

Reaction	Rate coefficient (<i>k</i>)	<i>k</i> at 250 K coefficient	Uncertainty in <i>k</i> (250 K)	255 K/250 K ratio	300 K/250 K ratio
OH production					
O(¹ D) + H ₂ O → 2OH	1.63e-10 exp(60/ <i>T</i>)	2.07E-10	1.09	1.00	0.96
NO + HO ₂ → NO ₂ + OH	3.44e-12 exp(260/ <i>T</i>)	9.72E-12	1.10	0.98	0.84
NO + CH ₃ O ₂ → CH ₂ O + NO ₂ + HO ₂	2.80e-12 exp(300/ <i>T</i>)	9.30E-12	1.23	0.98	0.82
HO ₂ + O ₃ → OH + 2O ₂	1.00e-14 exp(−490/ <i>T</i>)	1.41E-15	1.21	1.04	1.39
OH loss					
CH ₄ + OH → CH ₃ O ₂ + H ₂ O	2.45e-12 exp(−1775/ <i>T</i>)	2.02E-15	1.17	1.14	3.27
CO + OH → H + CO ₂	^a TROEE(<i>T</i> , <i>C</i>)	2.37E-13	1.07	0.97	0.96
NO ₂ + OH → HNO ₃	^b TROE(<i>T</i> , <i>C</i>)	1.21E-11	1.71	0.97	0.78
CH ₂ O + OH → CO + H ₂ O + H	5.50e-12 exp(125/ <i>T</i>)	9.07E-12	1.19	0.99	0.92
CH ₃ OOH + OH → 0.7*CH ₃ O ₂ + 0.3*OH + 0.3*CH ₂ O + H ₂ O	3.80e-12 exp(200/ <i>T</i>)	8.46E-12	1.54	0.99	0.88
CH ₃ OH + OH → HO ₂ + CH ₂ O	2.90e-12 exp(−345/ <i>T</i>)	7.30E-13	1.14	1.03	1.26
C ₃ H ₈ + OH → products	9.19e-12 exp(−630/ <i>T</i>)	1.75E-12	1.04	1.03	1.28
Isoprene + OH → products	3.0e-11 exp(360/ <i>T</i>)	1.26E-10	1.14	0.97	0.79

$$^a\text{TROEE}(T, C) = 1.85e - 13 e^{\frac{-65}{T}} \left(1 - \frac{\frac{\frac{6.9e-33 \left(\frac{298}{T}\right)^{2.1} C}{1.1e-12 \left(\frac{298}{T}\right)^{-1.3}} \times 0.6}{1 + \frac{6.9e-33 \left(\frac{298}{T}\right)^{2.1} C}{1.1e-12 \left(\frac{298}{T}\right)^{-1.3}}}}{\frac{1}{1 + \log_{10} \left(\frac{\frac{6.9e-33 \left(\frac{298}{T}\right)^{2.1} C}{1.1e-12 \left(\frac{298}{T}\right)^{-1.3}} \right)^2}} \right) + \frac{6.9e-33 \left(\frac{298}{T}\right)^{2.1} C}{1 + \frac{6.9e-33 \left(\frac{298}{T}\right)^{2.1} C}{1.1e-12 \left(\frac{298}{T}\right)^{-1.3}}} \times 0.6 \left(\frac{1}{1 + \log_{10} \left(\frac{\frac{6.9e-33 \left(\frac{298}{T}\right)^{2.1} C}{1.1e-12 \left(\frac{298}{T}\right)^{-1.3}} \right)^2} \right)$$

$$^b\text{TROE}(T, C) = \frac{1.8e-30 \left(\frac{298}{T}\right)^3 C}{1 + \frac{1.8e-30 \left(\frac{298}{T}\right)^3 C}{2.8e-11}} \times 0.6 \left(\frac{1}{1 + \log_{10} \left(\frac{1.8e-30 \left(\frac{298}{T}\right)^3 C}{2.8e-11} \right)^2} \right)$$

C in TROEE and TROE refers to air density; the rates are calculated at *C* = 2e19 molec cm^{−3}. Reaction rate coefficients and their respective uncertainties (expressed as multiplicative factors, calculated for 250 K) are from Jet Propulsion Laboratory Evaluation Number 19 (Burkholder et al. 2019). Also shown are ratios of the kinetic reaction rate coefficients for a +5 warming of the troposphere (255 K/250 K) and for free tropospheric versus warm surface air (300 K/250 K).

Compensating production and loss pathways create a strongly buffered chemical system in which OH does not respond proportionally to changes in its drivers, except for decreases in stratospheric ozone columns (Spivakovsky et al. 2000, table 6). Lelieveld et al. (2016) include new understanding of recycling via RO₂ chemistry and estimate a global OH recycling

Airmass-weighted

OH: a metric for global mean tropospheric OH simulated in atmospheric chemistry models that weights the OH concentration by the mass of air in each model grid cell; commonly reported in units of molecules cm^{-3} (the quantity needed to calculate reaction rates); other weighting conventions include weighting by volume or by $k_{\text{CH}_4+\text{OH}}(T)$

probability of 67%. This OH recycling is tied both to OH sinks (**Figure 1**) and to ozone production reactions ($\text{NO} + \text{HO}_2$ and $\text{NO} + \text{RO}_2$) that generally dominate OH recycling in the tropics and northern mid-latitudes (Lelieveld et al. 2016) (**Figure 2**). Overall, secondary OH formation exceeds primary production in continental source regions and at higher latitudes (Lelieveld et al. 2016). Even if global mean OH is stable, the regional balance of tropospheric oxidation processes could change with climate, with implications for the lifetimes of species lost by reaction with OH.

2.2. Decadal Trends

Decadal trends in OH, inferred from inverse modeling approaches constrained by long-lived gases that react with OH, or simulated with forward atmospheric chemistry models, are reviewed elsewhere (Chua et al. 2023, Szopa et al. 2021, Turner et al. 2019). We focus here on studies applying models to parse individual budget terms contributing to decadal trends in OH. Chua et al. (2023) find a 14% increase in total OH production and loss, and a 5% increase in global annual airmass-weighted OH (Lawrence et al. 2001) during 1980–2014, agreeing with the sign of simulated changes in other models (Stevenson et al. 2020, Turner et al. 2017). They attribute this increase to larger production driven by the concomitant rise in anthropogenic NO_x emissions and methane concentrations, which together increase ozone and associated primary production plus recycling via $\text{HO}_2 + \text{O}_3$ and $\text{HO}_2 + \text{NO}$. Over a similar time period (1980–2010), Zhao et al. (2020) find a multi-model average increase of $\sim 9\%$ in total OH production [sum of $\text{O}^1\text{D} + \text{H}_2\text{O}$ ($+5\%$), $\text{NO} + \text{HO}_2$ ($+3\%$), and $\text{O}_3 + \text{HO}_2$ ($+1\%$)] after interannual variability is removed. Observational and model analysis points to a role for increasing atmospheric water vapor in driving OH trends and variability in recent decades by enhancing primary production (Chua et al. 2023; Nicely et al. 2020, 2018). Primary production increases from 1988 to 1997 have been tied to the eruption of Pinatubo in 1991, which depleted stratospheric ozone and increased tropospheric ultraviolet radiation (Duncan & Logan 2008). Losses of OH generally increase to balance production, although Zhao et al. (2020) note a greater than 50% decline in the $\text{CO} + \text{OH}$ sink between 2000 and 2010, after removing interannual variability from biomass burning emissions.

2.3. Spatial Variations

Observations of well-mixed methyl chloroform and other proposed long-lived proxies for OH (Liang et al. 2017) constrain global annual mean OH (Patra et al. 2021) but cannot map spatial variations in OH beyond the interhemispheric gradient (Patra et al. 2014). Measurements of ^{14}CO may offer new constraints on spatial and seasonal variability in OH (Petrenko et al. 2021). Spivakovsky et al. (2000) developed a global-scale seasonal OH climatology by constraining a box model with measurements of OH sources and sinks. It has been used for more than two decades to benchmark global model simulations (e.g., Zhao et al. 2019). We briefly review promising applications of newly available satellite retrievals and airborne observations to map spatial patterns in OH.

Pioneering applications of satellite data show promise for mapping spatial variations in total or partial tropospheric columns of OH. Some studies use species that mirror OH production or loss under certain conditions: methane (Zhang et al. 2018), formaldehyde (HCHO) (Wolfe et al. 2019), and HCHO together with isoprene (Wells et al. 2020), although in situ sampling suggests large local variations may limit accuracy (Guo et al. 2023). Some studies use multiple chemical species retrieved from satellite to infer OH variations. For example, a machine learning approach reveals OH trends ranging from -17% to $+11\%$ in different North American cities due to anthropogenic emission changes (Zhu et al. 2022). A simplified steady-state model for OH production and loss

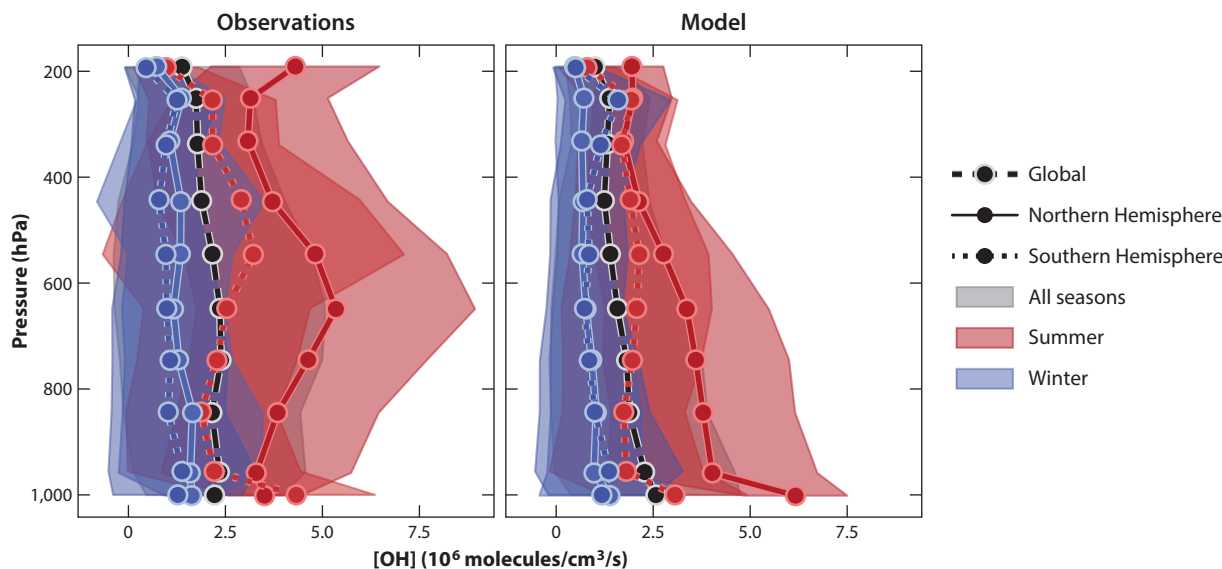


Figure 3

Measured and modeled OH concentrations are larger in the summer when radiation and water vapor are more abundant. The panels show vertical profiles of mean (*lines*) OH concentrations and standard deviation (*shaded polygons*) within pressure intervals of approximately 100 hPa measured during the ATom aircraft campaign (*left panel*, from 10-s samples) (Thompson et al. 2022) and modeled with the Geophysical Fluid Dynamics Laboratory's Atmospheric Model 3 chemistry-climate model, sampled at the nearest hour along the ATom flight tracks (*right panel*) (Baublitz et al. 2023). The data or model fields are extracted for climatological summer (*red*) or winter (*blue*) for the Northern (*solid lines*) or Southern Hemisphere (*dashed lines*). Hemispheres are averaged across all four seasons (and respective ATom campaigns) for the full ATom data set (Global, All seasons). The summer and winter data include the first and second ATom campaigns, ATom-1 and ATom-2, which spanned July–August 2016 and January–February 2017, respectively. Abbreviation: ATom, Atmospheric Tomography.

links interannual variability of tropical lower tropospheric OH (between -7% and $+8\%$) to the influences of ozone on primary production and of CO on OH loss (Pimlott et al. 2022). Satellite HCHO columns show an order-of-magnitude depression in OH over the tropical Pacific Warm Pool (Wolfe et al. 2019) where field measurements suggest a strong sensitivity to NO_x (Nicely et al. 2016).

The NASA Atmospheric Tomography (ATom) mission (2016–2018) obtained an unprecedented suite of measurements by profiling the remote marine troposphere around the globe in each season (Thompson et al. 2022). ATom OH measurements (Brune et al. 2020) show the OH dependence on radiation, with higher abundances in the summer hemisphere (**Figure 3**). A global atmospheric chemistry model sampled as consistently as possible with these measurements broadly captures the all-campaign-seasons average and seasonal hemispheric variations (**Figure 3**). The model, however, overestimates near-surface OH in summer, simulates smaller differences between the hemispheres during winter, and misses the mid-tropospheric peak evident in the summertime observations (**Figure 3**). A simplified OH steady-state model comprising $\text{O}(^1\text{D}) + \text{H}_2\text{O}$, $\text{NO} + \text{HO}_2$, and the reactivity of CO against OH captures much of the spatial variance in OH over the ATom flight tracks (Baublitz et al. 2023). More detailed chemical mechanisms reflect HO_x concentrations in the remote marine troposphere to within uncertainties of 40% (Brune et al. 2020).

A new approach (Prather et al. 2017, 2018) configures chemistry-climate models to operate as if each grid cell were a chemical box model constrained by ATom data and maps the resulting reaction rates for ozone loss (primary OH production), methane loss (OH sink), and ozone

OH reactivity:

the sum of all species serving as OH sinks, weighted by their reaction rate coefficients with OH: $\sum_{i=1}^N k_{x_i+\text{OH}}[x_i]$; the inverse lifetime of OH in units of s^{-1}

production (OH recycling via $\text{NO} + \text{HO}_2$ and $\text{NO} + \text{RO}_2$) along the flight tracks. A key finding is that certain air parcels dominate the chemistry controlling OH, with 80–90% of the total chemical processing via these reactions occurring in only 50% of air parcels, and at least a quarter of the processing occurring in only 10% of parcels (Guo et al. 2023), with a tight coupling through OH between ozone and methane loss rates (Prather et al. 2018). A sensitivity analysis using ATom data attributes OH spatial variations to water vapor fluctuations throughout much of the marine free troposphere (Baublitz et al. 2023). The 10-s ATom data further show that the chemical reactions controlling atmospheric oxidation by OH occur at scales resolvable by global models (Guo et al. 2023).

2.4. Uncertainties

Knowledge gaps in the processes controlling OH concentrations are particularly large in some atmospheric regions. Upper tropospheric model-measurement discrepancies (Brune et al. 2020) may reflect a missing NO_x source (Dang et al. 2023) and/or errors in NO – NO_2 – O_3 photochemistry (Silvern et al. 2018). Evaluation of current global models over the remote marine troposphere shows that they underestimate ozone production below 2 km where NO_x concentrations are too low (Guo et al. 2023), implying errors in OH recycling. Models also underestimate near-surface OH reactivity, possibly associated with oceanic emissions of organic gases (Thames et al. 2020, Travis et al. 2020). Other field measurements confirm known chemistry in the clean remote marine boundary layer but diverge in air masses containing aged pollution, implying the presence of additional RO_2 (Andersen et al. 2022). Overall, observational constraints on NO_x distributions would improve modeling of spatial variations in OH (Fiore et al. 2008, Labrador et al. 2004).

Uncertainty in global mean OH is apparent from the wide diversity of modeled trends and from the $\sim 50\%$ range in simulated global annual mean OH abundance (Murray et al. 2021) (Figure 4).

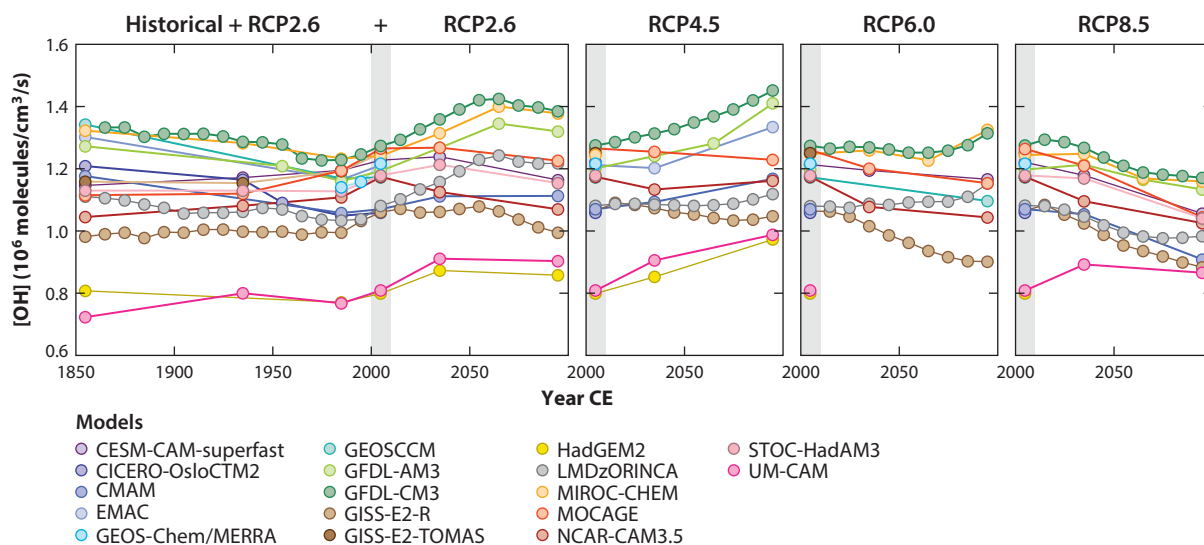


Figure 4

Global atmospheric chemistry models (colors; named in legend) differ widely in both the sign and magnitude of past and future changes in global decadal mean tropospheric OH even under the same historical (1850–2005) and future (four RCP scenarios from 2005 to 2100) anthropogenic emissions. Figure adapted with permission from Murray et al. (2021, figure 1). Abbreviation: RCP, representative concentration pathway.

Table 2 Uncertainties in OH abundance in the modern atmosphere as quantified in recent studies, classified into seven categories by process

Process	Findings	Reference
Heterogeneous (gas phase-aerosol) chemistry	HO ₂ uptake by aerosol reduces global mean airmass-weighted OH by 12%	Mao et al. 2013a
	Particulate nitrate photolysis increases global tropospheric OH concentrations by 19%	Shah et al. 2023
Photolysis and kinetic reaction rates	$J_{[\text{NO}_2]}$ and $k_{[\text{NO}_2][\text{OH}]}$ lead to uncertainty in OH by 20% and 30%, respectively, along the flight tracks in the Arctic Research of the Composition of the Troposphere from Aircraft and Satellites B field campaign	Christian et al. 2017
	Photolysis and kinetic rate coefficients, especially the NO ₂ + OH reaction rate, contribute to OH uncertainty of 20–30% along the flight tracks in the Intercontinental Chemical Transport Experiment-North America field campaign	Christian et al. 2018
	Kinetic uncertainties across 60 reactions yield an overall uncertainty on annual mean tropospheric OH of 17%	Newsome & Evans 2017
Weather-sensitive emissions	Uncertainty in lightning, surface NO _x emissions, and isoprene emissions contributes to 20%, 14%, and 10%, respectively, variations in modeled global mean OH	Wild et al. 2020
Fate of isoprene oxidation products	Uncertainty in isoprene emission and fate lead to global mean OH changes ranging from –8% to +17% from the preindustrial era to present day	Achakulwisut et al. 2015
	Improving the representation of isoprene oxidation decreases global annual mean tropospheric OH by 11%	Bates & Jacob 2019
NO _x fate	Different model treatments of the fate of reactive nitrogen and the oxidative efficiency of non-methane volatile organic compound lead to intermodel variations in global mean OH of 30%	Murray et al. 2021
Tropospheric halogen chemistry	The addition of tropospheric halogens leads to an OH decrease of 8%	Caram et al. 2023
	Halogen chemistry acts to decrease global mean OH by 4.5%	Stone et al. 2018
	Halogen chemistry decreases global tropospheric OH by 4%	Wang et al. 2021
	Natural halogens deplete approximately 13% of tropospheric O ₃ , leading to a reduction of OH	Iglesias-Suarez et al. 2020
Water vapor distributions	Uncertainty in humidity contributes to 20% variations in modeled global mean OH	Wild et al. 2020

The true uncertainty may be larger as model mechanisms adopt similar expert-recommended parameters to drive photolysis and kinetic reactions (e.g., Burkholder et al. 2019) and often, as in **Figure 4**, use the same anthropogenic emissions. Conclusive attribution of methane trends requires accurate trends in global mean OH to within a few percent, much smaller than the estimated uncertainty of about 7% (Turner et al. 2019). Efforts to ascribe observed methane variations to specific drivers are thus strongly method dependent (Turner et al. 2017).

Several studies have identified specific processes contributing to uncertainty in global mean OH abundance (**Table 2**). The intermodel OH differences in **Figure 4** have been linked to discrepancies in the oxidation efficiency of reactive carbon and in NO_x lifetimes and NO_y partitioning (Murray et al. 2021). A study probing sources of uncertainty in individual models underscores major knowledge gaps in the sensitivity of OH to water vapor and emissions of isoprene and NO_x (Wild et al. 2020). A key uncertainty involves the fate of isoprene, which is emitted by vegetation and undergoes a complex cascade of oxidative processes, some of which may produce, destroy, or regenerate OH (Achakulwisut et al. 2015, Bates & Jacob 2019). Major uncertainty also surrounds the role of aerosols as both radical sinks (Mao et al. 2013a) and sources (Shah et al. 2023). Halogen

chemistry, involving bromine, chlorine, and iodine reactions, may alter the relative importance of the OH production and loss pathways shown in **Figures 1** and **2**, but the sign and magnitude of this impact are likely species dependent, with knowledge still evolving rapidly (Caram et al. 2023, Stone et al. 2018, Wang et al. 2021). Intermodel differences in the sensitivity of OH to specific variables have been linked to atmospheric chemistry as represented within chemical mechanisms (Nicely et al. 2020). A formal Monte Carlo and addition-in-quadrature uncertainty analysis of reported uncertainties in kinetic reaction (and photolysis) rate coefficients estimates 16% uncertainty in global mean OH, with the reaction $\text{NO}_2 + \text{OH} + \text{M} \rightarrow \text{HNO}_3 + \text{M}$ (>50% uncertainty in **Table 1**) contributing most to the total uncertainty (Newsome & Evans 2017).

While the uncertainty in the seven categories of processes in **Table 2** is not well quantified, if we assume that each category—i.e., heterogeneous (gas phase-aerosol) chemistry, photolysis and kinetic reaction rates, weather-sensitive emissions, fate of isoprene oxidation products, NO_x fate, halogen chemistry, water vapor distribution—contributes 10–15% uncertainty, then addition in quadrature yields a total uncertainty of ~25–40%. Global sensitivity analysis including uncertainty in kinetics and photolysis as well as emissions, meteorology, and heterogeneous chemistry identifies regional and seasonal variations in the processes dominating the overall uncertainty in model OH sampled along flight tracks from airborne campaigns, with modeled and measured OH often agreeing to within their respective uncertainties (Christian et al. 2017, 2018). These uncertainty estimates (**Table 2**), however, are sufficiently large as to hinder definitive assessments of model skill in representing the balance among competing production and loss processes.

3. SENSITIVITY OF OH TO FACTORS MODULATED BY CLIMATE

In the modern atmosphere, global mean OH is sensitive to a limited set of factors (Section 2): ratios of emissions (or abundances) of reactive nitrogen and carbon (Wang & Jacob 1998), water vapor, tropospheric ozone, and ozone photolysis rates [$\text{JO}_3 \rightarrow \text{O}^1\text{D}$, which in turn depend strongly on overhead O_3 columns (e.g., Murray et al. 2014)]. Below, we summarize processes through which OH production and loss may respond to climate change.

3.1. Temperature-Sensitive Chemistry

One of the most robust relationships between climate variations and OH involves water vapor. Atmospheric water vapor increases by 7% per 1 K warming of global annual mean surface temperature (e.g., Held & Soden 2006). Where primary OH production is limited by the availability of water vapor, such as in the dry middle-to-upper troposphere, the overall HO_x supply will increase as the troposphere warms. A reduced-form model analysis over the remote marine troposphere attributes observed OH spatial variations to fluctuations in measured H_2O (Baublitz et al. 2023). Analysis of global models with field measurements points to a role for water vapor in driving intermodel differences in OH (Prather et al. 2018). Further work is needed to determine whether drivers of spatial variations deduced from field campaigns also control interannual variability in global mean OH. The increase in water vapor with temperature is likely the dominant response of OH production in a warmer climate (e.g., Johnson et al. 1999).

We consider the temperature dependence of several key chemical pathways through which OH is produced or lost (**Table 1**). We contrast rate coefficients for a warm near-surface (300 K) versus free tropospheric (250 K) air mass and for a +5 K increase in tropospheric mean temperature from 250 to 255 K, within the range of model projections for future climate change (IPCC 2023). The most temperature-sensitive loss rate coefficient is $k_{\text{CH}_4+\text{OH}}$, which increases by 14% and more than 300% for a 5 versus 50 K temperature increase (**Table 1**). This temperature sensitivity explains why the lifetime of methane and its response to climate change depend on how OH will change in

the warm tropical lower troposphere. In contrast, the primary loss pathway for OH, reaction with CO, has little dependence on temperature (Table 1). While most major OH production pathways are not strongly sensitive to temperature, $k_{\text{HO}_2+\text{O}_3}$ proceeds 4% and 39% faster for increases of +5 and +50 K relative to 250 K, and $k_{\text{NO}+\text{HO}_2}$ and $k_{\text{NO}+\text{CH}_3\text{O}_2}$ proceed 2% and more than 15% more slowly, respectively. While tropospheric warming could thus lead to small regional shifts in the balance of the recycling pathways shown in Figure 2, none of the changes for a +5 K tropospheric warming exceed the uncertainty in the reaction kinetics, highlighting the importance of continued laboratory efforts to narrow this uncertainty (Table 1).

3.2. Weather-Sensitive Emissions

Numerous gases serving as sources or sinks for OH are emitted naturally or as by-products of human activities. Figure 5 illustrates how these emissions may respond to a warmer climate and also how they influence OH. Weather-sensitive emissions are expected to increase with climate change, but the net impact on OH will depend on the balance of reactive carbon (methane, CO, NMVOCs; OH sinks in Equation 2) to NO_x (OH source in Equation 1) and will vary regionally. Below, we review the sensitivity of these sources (lightning, biogenic, wildfires, anthropogenic) to meteorology and their role in producing or destroying OH.

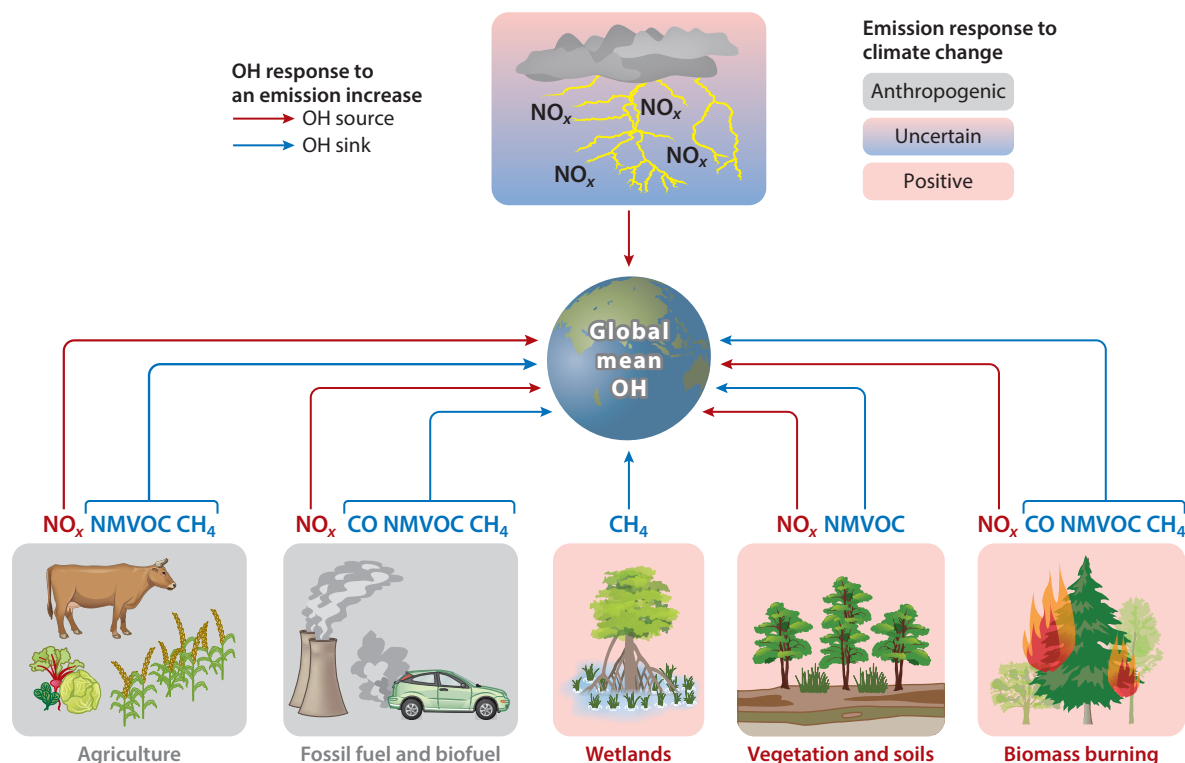


Figure 5

Schematic illustrating how emissions of trace gases serve as sources (red arrows) or sinks (blue arrows) of OH, and the anticipated response of those emissions to climate change: positive (pink shading) for biogenic emissions of methane from wetlands, NMVOC from vegetation, NO_x from soils, and biomass burning, or uncertain (pink-to-blue shading) for lightning NO_x . Future changes in anthropogenic emissions (gray shading) will most likely be dominated by societal decisions and not by climate change. Abbreviation: NMVOC, non-methane volatile organic compound.

LIGHTNING HO_x

Lightning storms have long been hypothesized to produce HO_x, in addition to NO_x (Section 3.2), but early studies suggested that the lifetime of lightning HO_x was too short to influence global oxidative capacity (Bhettanabhotla et al. 1985). Recent aircraft measurements, however, show that lightning storms generate OH abundances several orders of magnitude greater than previously thought, potentially contributing 2–16% of global OH oxidation capacity (Brune et al. 2021). In addition to visible lightning, electrified storms contain weaker electrical discharges, such as streamers and corona (Kostinskiy et al. 2020), some of which are subvisible. These weaker discharges generate little heating but produce ultraviolet radiation or electrons with sufficient energy to dissociate both water vapor and molecular oxygen, leading immediately to production of OH and ozone (Jenkins et al. 2021). A model study suggests that a direct lightning source of OH could enhance mass-weighted global OH by 0.3–10%, with the large range of uncertainty indicating the need for further process-based observational constraints (Mao et al. 2021). Quantifying the ozone produced in electrical storms is difficult because it cannot easily be distinguished from the ozone-rich stratospheric air that is pulled into the troposphere during these storms (Brune et al. 2021).

Lightning NO_x is one of the strongest levers by which OH increases globally, given the lack of coemitted species serving as OH sinks (Murray et al. 2013) (**Figure 5**). New evidence suggests lightning may substantially boost the atmospheric oxidizing capacity by producing HO_x (see the sidebar titled Lightning HO_x). The extremely high temperatures that characterize lightning flashes initiate a series of reactions leading to large quantities of NO_x (Murray 2016, Navarro-González et al. 2001, Pickering et al. 1990). Lightning NO_x enhances ozone globally, with tropical lightning influencing the chemical composition even over mid-latitudes (Mickley et al. 2001). Global atmospheric chemistry-climate models commonly parameterize this source as a power-law dependence to convective cloud top heights, which usually, though not always, increases with climate warming [−0.03 to 0.61 Tg(N) year/K (Thornhill et al. 2021)], or about 10% globally for every +1 K (Price 2009). A parameterization based on upward cloud ice flux, linked more closely to microphysics, instead suggests climate warming decreases the global flash rate and associated NO_x source (Finney et al. 2018) [15% decrease under a strong warming [representative concentration pathway (RCP) 8.5] scenario]. In a cloud-resolving model, Roms (2019) probes different proxies for flash rates, including a scaling for lightning flashes based on convective available potential energy and precipitation, but is unable to conclusively determine the sign of the lightning response to climate change, which is likely to vary by region (e.g., Chakraborty et al. 2021, Chen et al. 2021, Kahraman et al. 2022).

Overall, biogenic sources are expected to increase in a warmer climate (**Figure 5**). These sources include methane produced by methanogens in wetlands, BVOC emitted from vegetation, and NO produced during microbial nitrification or denitrification within soils. A positive relationship between CO₂ and wetland methane emissions is supported by ice core records (Petit et al. 1999), although the magnitude of this relationship is uncertain and model dependent (Melton et al. 2013). Process-based wetland models tend to simulate declining tropical methane emissions with higher temperatures and increasing global emissions with precipitation, but these dependencies are uncertain (Melton et al. 2013). Climate-driven changes in wetland emissions are neglected in most chemistry-climate multi-model studies to date, but process-based wetland models suggest increases of 33–60% by 2100 (Dean et al. 2018), with two estimates from chemistry-climate models indicating 20 or 60 Tg CH₄ year/K (Thornhill et al. 2021). Additional wetland methane emissions with rising CO₂ would substantially enhance atmospheric methane (Kleinen et al. 2021), increasing the OH sink.

Chemistry-climate models that allow BVOC emissions to respond to meteorology (Guenther et al. 2012) generally project future increases with temperature [e.g., 30–230 Tg year/K (Thornhill et al. 2021)] and thus increase the OH sink (**Figure 5**). While rising atmospheric CO₂ may offset some of the expected temperature-driven increase in isoprene (the most abundant individual BVOC), the magnitude of this effect may be lessened as temperatures rise (Sharkey & Monson 2014). Soil NO emissions also increase with temperature, and pulsing occurs following precipitation on dry soils (Hudman et al. 2012). Increasing soil NO_x with global warming would enhance OH and ozone (**Figure 5**). Higher sea surface temperatures under climate change may increase the flux of marine halogens, produced by either biochemical or abiotic processes in the ocean into the atmosphere in some regions, with uncertain consequences for ozone and in turn OH (Iglesias-Suarez et al. 2020).

Wildfires are strongly sensitive to rising temperature and drier conditions (Abatzoglou & Williams 2016), and climate change is expected to increase wildfire activity in many regions (Jones et al. 2022). All of the gases in **Figure 5** are emitted by fires and serve as both sources and sinks of OH. Observations provide evidence for OH production locally within smoke plumes (Theys et al. 2020), but models suggest that the sink gases dominate the net impact on OH in the present global atmosphere (Murray et al. 2013) and that a doubling of wildfire emissions could decrease global mean OH by about 6% (Mao et al. 2013b).

Anthropogenic emissions are usually assumed to have little sensitivity to temperature relative to the other sources discussed above (**Figure 5**). Some anthropogenic emissions increase on warmer days, including evaporative NMVOC and agricultural emissions. In some regions, power plant emissions also increase with cooling demand on warmer days (Abel et al. 2017). Independent of climate change, continued increases in anthropogenic NO_x from aircraft and ships will increase OH (Butler et al. 2020, Terrenoire et al. 2022), although estimated OH increases may be biased high due to poor resolution of plume chemistry in global models (Eastham & Jacob 2017). Given their competing influences on production and loss pathways, the net influence of weather-sensitive emissions on OH is highly uncertain. For example, if lightning NO_x were to decline while wetland methane emissions increased, then the global OH abundance would decrease in response.

3.3. Climate Variability and Possible Dynamical Changes

Even though locally at a steady state, OH concentrations fluctuate as internal climate variability (Lehner & Deser 2023) causes variations in OH production and loss pathways through the processes discussed above. **Figure 6** illustrates that 5–10% variations (one standard deviation of the range across an ensemble generated with a single model) in annual mean tropospheric column OH can occur solely because of climate variability. This range is similar to year-to-year variability of generally less than ±5% simulated by other global atmospheric chemistry models (Szopa et al. 2021), with variations as high as 2% linked to the net effects of meteorology and biomass burning alone (Chua et al. 2023). Two studies suggest that interannual OH variability has weakened in the early twenty-first century versus the last few decades of the twentieth century (He et al. 2020, Zhao et al. 2019). Murray et al. (2013; see their table 2) quantify the sensitivities of interannual variability in global annual mean OH to specific production and loss pathways, and they highlight a strong sensitivity of OH to lightning NO_x over the 1998–2006 period.

El Niño–Southern Oscillation (ENSO) is the dominant mode of naturally arising climate variability (Lee et al. 2021) and is known to influence global OH. This strong regional signal in the tropical Pacific is evident in **Figure 6**. During the boreal winter season, ENSO modulates approximately 30% of the total variance in tropospheric column and upper tropospheric OH (Anderson et al. 2021). This influence occurs through multiple production and loss pathways involving water

Internal climate variability: climate noise; fluctuations arising naturally in Earth's atmosphere and through climate system couplings such as with the ocean, of which a prime example is the El Niño–Southern Oscillation in the tropical Pacific Ocean

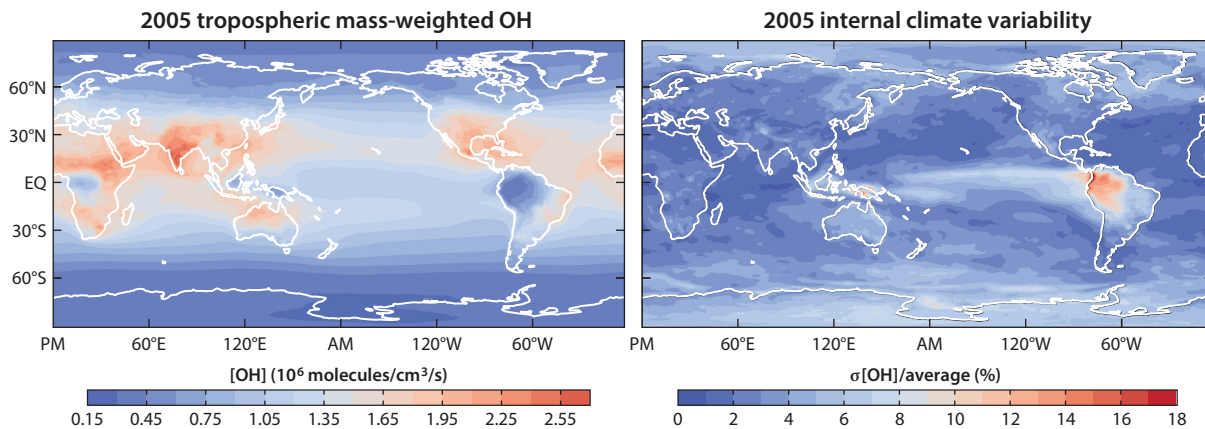


Figure 6

Spatial distribution of tropospheric mean airmass-weighted column OH (*left*) and variations arising solely from internal climate variability (*right*). These estimates are based on a 13-member initial-condition ensemble available with full tropospheric and stratospheric chemistry (Fiore et al. 2022) in which all forcings (e.g., greenhouse gases and air pollutants) are identical in each ensemble member. Differences in ensemble members arise from different initial conditions, the use of which reflects only a tiny perturbation to atmospheric temperature for some members at the beginning of the simulation (January 1, 1950) (see Deser et al. 2012). Specifically, the left panel shows the annual tropospheric mean airmass-weighted column OH in 2005 averaged over all ensemble members, and the right panel shows the standard deviation (σ) across the ensemble members divided by the ensemble mean.

vapor and tropospheric ozone (Chandra et al. 1998, Ziemke et al. 2010), lightning NO_x (Turner et al. 2018), and biomass burning (Duncan et al. 2003). Regional OH responses occur through atmospheric teleconnections involving changes in upper tropospheric NO_x and lower tropospheric primary production (Anderson et al. 2021). Models indicate net negative OH anomalies during El Niño events due to biomass burning (e.g., Zhao et al. 2020), with tropospheric column OH as much as 10–15% lower over the tropical Pacific and Australia (Anderson et al. 2021). Higher OH during La Niña has been attributed to enhanced lightning NO_x (Anderson et al. 2021, Turner et al. 2018). Despite the lack of consensus on future changes in ENSO intensity or variability, ENSO will continue to dominate interannual variability (Lee et al. 2021), amplifying (e.g., the 2023 El Niño) or offsetting the overall upward warming trend in any given year. This variability will thus continue to influence OH globally and should be accounted for when interpreting observed trends in OH as well as the trace gases whose lifetimes are set by OH.

Climate change may also influence OH by changing atmospheric circulation. As water vapor rises, convective mass flux decreases (Held & Soden 2006), potentially altering lightning NO_x and the vertical mixing of OH source and sink gases. An expanding tropical area under climate warming may increase primary production of OH (Nicely et al. 2018). Circulation changes can alter mixing between the tropics and higher latitudes, affecting the lifetimes of species, such as methane, which is primarily lost with OH. Trends in stratospheric circulation patterns (Abalos et al. 2020, Hegglin & Shepherd 2009, Neu et al. 2014) also affect OH by changing two terms for primary production: stratosphere-to-troposphere ozone transport and ultraviolet radiation penetrating into the troposphere.

4. PREINDUSTRIAL AND PALEO OXIDATION CAPACITY

Gases and aerosols trapped in ice cores in polar regions or high-altitude tropical glaciers can serve as proxies to reconstruct concentrations of oxidative gases in preindustrial and paleo atmospheres (Alexander & Mickley 2015). Lessons learned from the past can be used to constrain model

efforts of future atmospheres. For example, knowledge of how the lower temperatures during the last glacial period affected the flux of stratospheric ozone into the troposphere would increase confidence in model simulations of future changes in this flux (Wang et al. 2022). Most oxidants, including OH, are not preserved directly in ice cores. In this section, we describe proxies for past oxidants and discuss modeling of preindustrial and paleo atmospheres. Following Hawkins et al. (2017), we define the preindustrial era from 1720 to 1800; the last glacial period extends from 115,000 to 11,700 years ago.

4.1. Paleo and Preindustrial Observations

Hydrogen peroxide (H_2O_2 or HOOH) is the only oxidative component of the atmosphere that has been directly measured in ice cores. HOOH is a reservoir species for HO_x (Figure 2), so its abundance provides clues on past OH variations (Anderson et al. 2022), especially in combination with models (e.g., Murray et al. 2014). The Antarctic ice core record indicates at least a 50% increase in HOOH since 1900, with most of the increase occurring since 1970 (Lamarque et al. 2011). Although ice core measurements of HOOH exist as far back as 1500 (Frey et al. 2006), interpretation of these records is complicated by chemical processing of this species within the ice. Nonetheless, the recent 1900–2000 increase in observed HOOH is consistent with increasing tropospheric ozone. In addition, loss of stratospheric ozone over Antarctica may have allowed more ultraviolet sunlight to penetrate the troposphere and accelerate ozone photolysis, increasing OH and HOOH (Lamarque et al. 2011).

Three other gases preserved in ice cores may indirectly shed light on the oxidative capacity of the atmosphere: HCHO , CO , and methane. HCHO is a product of the oxidation of NMVOCs and methane, which can serve as an OH sink or instead photolyze to generate OH (Bates & Jacob 2019) (Figure 2). Like HOOH , HCHO can undergo postdepositional chemistry in ice, and so its ice core record must be interpreted with caution (Hutterli et al. 2004) although measurements suggest an increase since the preindustrial era. Increasing CO and methane, both sinks for OH (Figure 2), are expected to decrease OH. CO may also be affected by in situ chemical processing in ice, and its variations may reflect the decomposition of organic material, especially compounds emitted during biomass burning (Faïn et al. 2014).

In contrast to HCHO and CO , methane is stable in ice, and trends in ice core records closely match those in the atmosphere over time. These records reveal low levels of methane during the last glacial period compared to the preindustrial era, likely due to a reduction in wetland areas (Rhodes et al. 2017). During the cold, dry glacial period, wetlands in the tropics shrank while in boreal regions polar ice caps covered wetlands (Kaplan et al. 2006). During the Holocene, wetlands expanded in size and methane levels rose. Since the 1800s, methane has more than doubled, reflecting oil and gas systems, agricultural activity, and mining (e.g., Turner et al. 2016). As methane and the CO produced during its oxidation are sinks of OH, interpreting the long-term record of methane in ice can yield insights into past OH levels.

Isotopic signatures in ice core components also contain information on oxidation capacity in past atmospheres. Such components include molecular oxygen, methane, CO , and sulfate and nitrate aerosol. For example, studies have relied on an isotopic quantity, $\Delta^{17}\text{O}(\text{SO}_4^{2-})$, to probe oxidation processes in the last glacial period. Sulfate aerosol is produced via oxidation of sulfur species by HO_x , ozone, or HOOH , each of which records a different signature of $\Delta^{17}\text{O}(\text{SO}_4^{2-})$ (Alexander et al. 2004, Sofen et al. 2011). Values of $\Delta^{17}\text{O}(\text{SO}_4^{2-})$ from an Antarctic ice core suggest a greater abundance of atmospheric OH during the last glacial period compared to the present day, at least locally (Alexander et al. 2002). Measurements of Δ_{36} reveal the relative proportions of ozone generated in the troposphere versus in the stratosphere (Yeung et al. 2012). By tracking

$\Delta^{17}\text{O}(\text{SO}_4^{2-})$:

a measure of the anomalous enrichment in ^{17}O isotope relative to ^{18}O in the oxygen atoms of sulfate aerosol preserved in ice cores; given mass-dependent fractionation processes, the expected relationship of these isotopes is $\delta^{17}\text{O} \cong \delta^{18}\text{O} \times 0.5$, with δ denoting the ratios of these isotopes to ^{16}O ; mass-independent fractionation is defined as $\Delta^{17}\text{O} = ^{17}\text{O} - (0.512 \times ^{18}\text{O})$

Δ_{36} : a measure of the proportion of ^{18}O occurring in molecular oxygen, relative to a random distribution of these isotopes in oxygen; because ozone photochemistry in cold regions such as the lower stratosphere increases Δ_{36} , deviations in ice core Δ_{36} can provide clues on past sources of tropospheric ozone

Δ_{36} in ice cores and firn, Yeung et al. (2019) determined that global tropospheric ozone has likely increased by 40% at most since the preindustrial era.

4.2. Models of Paleo and Preindustrial OH

Three-dimensional chemical transport models have been deployed to recreate the oxidation capacity of past atmospheres and to interpret the observations described in Section 4.1. Some of these models feature fully interactive chemistry-climate capability, in which modeled changes in aerosols and greenhouse gases affect meteorology, which in turn influences atmospheric composition. Multi-model comparisons reveal little consensus on changes in OH abundance since the last glacial period or even since the preindustrial era (Alexander & Mickley 2015) (**Figure 4**).

In past atmospheres, models suggest that global mean OH was most sensitive to lightning NO_x emissions, water vapor, and the density of the stratospheric ozone layer (Murray et al. 2013, 2014). Trends in biomass burning likely had relatively little effect, as fire emissions include both nitrogen and carbon species, with opposing effects on OH abundance (**Figure 5**). During the last glacial period, models agree that lower global temperatures and water vapor likely depressed OH levels, with some compensation possible from lower biogenic emissions (Achakulwisut et al. 2015, Valdes et al. 2005). **Figure 4** shows that models differ not just in magnitude but also in the sign of their simulated OH trend from the preindustrial to the present. Some of this intermodel diversity is tied to different ratios in the global mean tropospheric CO to NO_x burdens (Naik et al. 2013).

5. OH IN THE TWENTY-FIRST CENTURY

Projection of OH into the twenty-first century is challenging due to an array of uncertainties that can be classified into three major categories (Hawkins & Sutton 2009). First, internal variability, inherent in the climate system, is an irreducible source of uncertainty (Deser et al. 2012, Lehner & Deser 2023) (**Figure 6b**). Second, model (structural) uncertainty is potentially reducible as knowledge advances and model representation of the governing processes improves (Hawkins & Sutton 2009). At present, intermodel differences (**Figure 4**) indicate insufficient knowledge of how OH will respond to changing meteorology and how different chemical and meteorological feedbacks on oxidation operate. Third, scenario uncertainty reflects the collective but unknown choices by individuals and governments and the resulting anthropogenic emissions that yield different oxidative capacities (**Figure 4**). For example, societal choices regarding energy and agricultural systems will affect OH source and sink gases (**Figure 5**), and thereby the balance among OH production and loss pathways (**Figures 2 and 3**).

Many atmospheric chemistry models forced with future climate and air pollutant emission scenarios project global annual mean OH changes by 2100 that are similar to within 20%, with the sign dependent both on the scenario and on the representation of model processes (Murray et al. 2021, Voulgarakis et al. 2013, Wild et al. 2020). Models robustly indicate that climate change alone will increase OH due to rising water vapor, which together with the strong temperature sensitivity of $k_{\text{CH}_4+\text{OH}}$ (**Table 1**) leads to a negative climate feedback by shortening the methane lifetime against OH by 0.33 ± 0.1 year/K (Voulgarakis et al. 2013). From the perspective of OH, the $k_{\text{CH}_4+\text{OH}}$ temperature sensitivity buffers OH increases from rising water vapor (John et al. 2012). This increase in the OH sink is amplified by the feedback of methane on its own lifetime because methane and the CO produced during its oxidation are major OH sinks (Prather 1996, Holmes 2018) (Section 2.1). To date, most chemistry-climate models do not include increases in methane emissions from wetlands in response to rising CO_2 (Section 3.2), a phenomenon that would further deplete OH. The net balance of these climate-sensitive processes also depends on future stratospheric ozone concentrations, which will respond to both climate change and

Table 3 Expected sign of the response of global tropospheric mean OH to changes in technology implemented to mitigate climate change

Technology	Intended climate response	OH response
Hydrogen gas replaces fossil fuel	Reduce CO ₂ emissions and corresponding radiative forcing (but could increase radiative forcing from CH ₄ , O ₃ , and stratospheric H ₂ O)	Decrease OH by increasing sink
Sulfate aerosol injection to the stratosphere	Reflect more solar radiation back to space (but could deplete stratospheric ozone layer)	Increase OH with higher ultraviolet penetration into troposphere
Marine cloud brightening with sea salt aerosol	Reflect more solar radiation back to space	Decrease OH due to lower ozone from increased loss by halogens
Solar/wind/nuclear replace fossil fuel	Reduce CO ₂ emissions and corresponding radiative forcing	Uncertain; OH should tend toward preindustrial conditions
Methane removal	Reduce atmospheric methane concentrations and associated radiative forcing	Increase OH as the sink to methane decreases
Carbon dioxide removal	Reduce atmospheric carbon dioxide concentrations and associated radiative forcing	Diminish OH changes in future climate; could increase OH source and sink gases from fossil fuel combustion, but net effect is uncertain

human-driven emissions of ozone depleting substances. In one model, instantaneous cessation of all anthropogenic methane emissions relative to a future scenario [socio-economic pathway (SSP) 3–7.0] decreases surface methane concentrations by more than 30% to below preindustrial concentrations, attributed partially to increased water vapor abundances (Staniaszek et al. 2022). Models, however, differ in both their simulations of water vapor (Prather et al. 2018) and OH sensitivity to water vapor (Wild et al. 2020).

5.1. Technological Wildcards

Novel technologies emerging during the twenty-first century intended to mitigate climate change could perturb atmospheric OH, and we briefly provide a few examples here as summarized in **Table 3**. Hydrogen (H₂) has been proposed as an alternative energy source to fossil fuel. H₂ burns cleanly, without carbon by-products that lead to the greenhouse gas CO₂. However, the production of H₂ may involve significant methane emissions (Bertagni et al. 2022, Ocko & Hamburg 2022). Once produced, H₂ can easily escape from pipelines and other infrastructure to the ambient atmosphere and compete with methane for reaction with OH (Equation 2). A longer methane lifetime would increase climate warming, as would the H₂-mediated increase in ozone production and stratospheric water vapor (Ocko & Hamburg 2022, Paulot et al. 2021).

Proposed initiatives for solar geoengineering could also alter tropospheric OH, with the sign dependent on the method. Injecting a layer of aerosols into the stratosphere to reflect incoming sunlight back to space and thus cool the planet (Irvine et al. 2019, MacMartin et al. 2019) could, depending on aerosol composition, diminish the abundance of stratospheric ozone through heterogeneous reactions (Pitari et al. 2014). The resulting increase in the flux of ultraviolet sunlight into the troposphere could ramp up OH primary production by increasing photolysis rates of tropospheric ozone. A typical scenario for stratospheric aerosol injection could increase global mean OH in the troposphere by as much as 9% (Moch et al. 2023).

A second technique of solar geoengineering, marine cloud brightening, involves artificially increasing the albedo, or reflectivity, of low-level clouds over tropical oceans. Injecting large quantities of sea salt aerosols would increase the number of tiny cloud droplets and enhance cloud brightness (Latham et al. 2012, MacMartin et al. 2018). As more incoming sunlight is reflected

Methane removal: technology or process enabling direct air capture of methane followed by its destruction (such as through oxidation); if deployed at sufficiently large scales, could lower atmospheric methane concentrations

Carbon dioxide removal: technology or process that converts atmospheric CO₂ to a stable reservoir for storage; if deployed at sufficiently large scales, could lower atmospheric CO₂ concentrations

back to space, the ocean surface below cools. However, the increased flux of sea salt aerosol into the atmosphere could potentially enhance the abundance of halogen species, with uncertain consequences for ozone and OH (Section 2). Applying a range of potential scenarios, one model study predicted that marine cloud brightening could decrease tropospheric ozone by 3–6% and global OH by 3–5% (Horowitz et al. 2020).

A future move away from dependence on fossil fuel combustion and toward nuclear energy and renewables such as solar and wind power as key sources of energy would presumably lead to emissions more consistent with the preindustrial atmosphere. New technologies for methane removal are receiving attention as a possible approach to mitigate climate (Jackson et al. 2021). OH concentrations should increase as the sink to methane declines, further reducing methane. Finally, carbon dioxide removal without any additional application of pollution control technologies would diminish future OH changes relative to an atmosphere with continued increases in CO₂, but OH could either increase or decrease in response to additional fossil fuel use.

SUMMARY POINTS

1. Tropospheric concentrations of the hydroxyl radical (OH) set the atmospheric lifetimes of some greenhouse gases (methane, ozone), air pollutants (CO, volatile organic compounds), and replacements for stratospheric ozone-depleting substances (hydrofluorocarbons and hydrochlorofluorocarbons), tying our capacity to predict future environmental change to knowledge of the processes controlling OH (**Figure 1**).
2. The most strongly temperature-sensitive chemical processes controlling OH production and loss are the dependence of OH primary production on water vapor and the kinetic rate coefficient for the methane-OH reaction (**Table 1**).
3. Under climate change, emissions of OH source and sink gases are expected to increase but the net impact on OH depends on the balance of the source from nitrogen oxides versus reactive carbon OH sinks (**Figure 5**).
4. Even with its atmospheric lifetime of a few seconds at most, OH is strongly modulated by internal climate variability, including through the El Niño–Southern Oscillation influence on meteorology and OH source and sink gases (**Figure 6**).
5. While modeled and measured OH often agree, at least within their uncertainty ranges of 5–30% (**Table 2**), major knowledge gaps remain in the regional and global balances among OH production and loss pathways, most notably in the marine boundary layer and upper troposphere where model-measurement discrepancies are largest.

FUTURE ISSUES

1. Future efforts should aim to reduce uncertainty in measurements, fundamental photolysis and reaction rates, and model process representation, including through coordinated laboratory, field, and modeling efforts.
2. Advances in observing systems and models will enable insights into OH variations at subannual to interannual scales, the processes controlling these variations including OH reactivity across different atmospheric environments (e.g., urban, forested, clean, fire

influenced, remote), and their importance in setting the lifetimes of key atmospheric trace gases.

3. Technological transitions proposed to mitigate climate change should account for OH and other oxidant responses to help avoid unintentional consequences.
4. Efforts to interpret proxies of oxidative species from ice cores and reconstruct oxidizing capacities in paleo atmospheres can help to illuminate how the atmospheric oxidizing capacity will respond to future climate change and to shifts in energy systems.
5. Agricultural sources are growing in magnitude and in relative importance as other anthropogenic emissions decline, yet their net impact on oxidizing capacity is unclear.
6. Additional lab and field studies are needed to constrain emerging knowledge of atmospheric chemistry, such as the role of aerosols in producing and destroying radicals, and of oxidation by halogens relative to OH.

DISCLOSURE STATEMENT

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